O-L-164D
January 16, 1970
SUPERSEDING
Fed. Spec. O-L-164C
April 5, 1965

FEDERAL SPECIFICATION

LEATHER DRESSING, MILDEW PREVENTIVE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. This specification covers two types of liquid preservatives suitable for leather dressing (see 6.1).

1.2 Classification.

1.2.1 Types. Leather dressing, preservative and mildew preventive compound, shall be of the following types, as specified (see 6.2).

Type I - Solvent.

Type II - Emulsifiable.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

O-T-236 - Tetrachloroethylene, (Perchloroethylene); Technical Grade,
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-P-704 - Pails, Metal (Shipping, Steel, 1 through 12 gallon).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 311 - Leather, Methods of Sampling and Testing.
Fed. Test Method Std. No. 536 - Soap and Soap-Products (Including Synthetic Detergents): Sampling and Testing.
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 Materials. The materials used in the leather dressing compound shall consist of a mixture of animal oil, mineral oil, fungicide, and solvents of such physical characteristics and in such proportions as to conform to the requirements of this specification.

3.1.1 Paranitrophenol. The paranitrophenol shall conform to the following requirements when tested as specified in 4.2.1.1:

	Minimum	Maximum
Paranitrophenol, percent	96.0	---
Melting point, °C.	111	114

3.1.2 Mineral oil. The mineral oil shall be a refined hydrocarbon oil and shall contain no additives. The oil shall conform to the following requirements when tested as specified in 4.2.1.1:

	Minimum	Maximum
Specific gravity 20° / 4°C.	0.885	0.924
Color, ASTM	-	3.5
Neutralization number	-	0.03
Viscosity, SU at 100°F. in seconds	135	160

3.1.3 Perchloroethylene. The perchloroethylene shall conform to O-T-236 for pH and free chlorine requirements only.

3.1.4 Cyclohexanone. The cyclohexanone shall conform to the following requirements when tested as specified in 4.2.1.1:

	Minimum	Maximum
Refractive index at 25°C.	1.4460	1.4520
Specific gravity, 25° /4°C.	0.940	0.950
Color, ASTM	-	0.5
Distillation range at 760 mm. Hg. pressure:		
5 mls	154.0°C	-
95 mls	-	157.0°C.
Water content, percent by weight	-	1.0

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3.1.5 Sulfonated cod oil. The sulfonated cod oil (see 6.3) shall be of a quality which will meet the detail end item requirements when formulated as specified in 3.2.2.

3.1.6 Sulfonated neatsfoot oil. The sulfonated neatsfoot oil (6.4) shall be of a quality which will meet the detail end item requirements when formulated as specified in 3.2.2.

3.2 Composition and chemical properties.

3.2.1 Type I. The leather dressing compound shall conform to the requirements specified in table I when tested as specified in 4.2.4.

TABLE I. Composition and chemical requirements for type I

Requirement	Minimum	Maximum
Composition (percent by weight):		
Paranitrophenol	1.8	2.2
Mineral oil	18.0	22.0
Perchloroethylene	66.0	70.0
Cyclohexanone	9.0	11.0
Chemical properties:		
Nonvolatile residue, percent	19.8	24.2
Unsaponifiable matter, percent	18.0	20.0
Specific gravity, 25°/4°C.	1.25	1.35
Matter insoluble in alcohol	None	None

3.2.2 Type II. The leather dressing compound shall include either a sulfonated cod oil or a sulfonated neatsfoot oil and paranitrophenol and shall conform to the requirements specified in table II when tested as specified in 4.2.4.

TABLE II. Composition and chemical requirements for type II

Requirement	Minimum	Maximum
Composition (percent by weight):		
Sulfonated cod oil or sulfonated neatsfoot oil	94.5	95.0
Paranitrophenol	5.0	5.5
Chemical properties:		
Water content, percent	22.6	24.7
Nonvolatile residue, percent	75.3	--
Organically combined SO_3, percent	3.3	5.2
Unsaponifiable nonvolatile matter, percent	--	1.9
Free fatty acid as oleic, percent	--	26.0
pH (acidity)	5.5	6.5
Matter insoluble in water	None	None

3.3 Labeling.

3.3.1 Unit containers. Each container of leather dressing compound shall be clearly and legibly lithographed or printed on the body with water-resistant black ink.

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3.3.1.1 Type I. Each container shall be marked as follows:

LEATHER DRESSING,

MILDEW PREVENTIVE COMPOUND

Solvent, Type I, Specification O-L-164

(Stock Number)

DO NOT USE ON LEATHER PRODUCTS WHICH COME

INTO PROLONGED CONTACT WITH THE SKIN

INGREDIENTS

Mineral oil	Percent by weight
Cyclohexanone	Percent by weight
Paranitrophenol	Percent by weight
Perchloroethylene	Percent by weight
100 percent	

Manufacturer's name
Contract number
Date
(Unit quantity)

DIRECTIONS FOR USE

Type I treating compound should be used as received (no dilution necessary). Wash the leather items with soap and water at a temperature not to exceed 100°F. A washing machine or other device in which leather items are tumbled may be used. Remove dirt remaining after this treatment by rubbing with a cloth or a brush. Rinse with clean water. Allow leather to drain for 5 minutes. Apply the compound by immersing the leather items for 2 minutes. When only a small quantity is treated, application may be made with a soft bristled brush, felt daubber, or cloth. Apply in a well-ventilated area using rubber gloves. After applying compound, wash hands thoroughly with cool water and soap. Allow leather items to dry thoroughly in the open air.
WARNING: Avoid skin contact or inhalation of fumes.

3.3.1.2 Type II. Each container shall be marked as follows:

LEATHER DRESSING,

MILDEW PREVENTIVE COMPOUND

Emulsifiable, Type II, Specification O-L-164

(Stock Number)

DO NOT USE ON LEATHER PRODUCTS WHICH COME

INTO PROLONGED CONTACT WITH THE SKIN

INGREDIENTS

Sulfonated cod oil or sulfonated neatsfoot oil	Percent by weight
Paranitrophenol	Percent by weight
100 percent	

Manufacturer's name
Contract number
Date
(Unit quantity)

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DIRECTIONS FOR USE

Type II treating compound is prepared by adding one gallon of the base material to five gallons of water.

If dirty, wash the leather items with soap and water at a temperature not to exceed 100°F. A washing machine or other device in which the leather items are tumbled may be used. Remove dirt remaining after this treatment by rubbing with a cloth or brush. Rinse with clean water. Allow to drain for 30 minutes. The compound should be applied on the basis of 80 pounds of leather items to 1 gallon of base material. Apply the compound by immersing the leather items and stir or tumble for 10 minutes. The residual solution should then be discarded and a new solution prepared.

When only a few items are to be treated, application of the undiluted compound may be made with a soft bristled brush, a felt dauber or cloth. User shall wear rubber gloves for application.

After applying the compound, wash hands thoroughly with cool water and soap. Allow leather items to dry thoroughly in the open air.

WARNING: Avoid skin contact or inhalation of fumes.

3.4 Workmanship. The leather dressing compound shall be a homogeneous liquid and shall conform to the quality established herein. The occurrence of defects shall not exceed the number permitted under the applicable acceptable quality level (AQL).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.1.1 Testing of materials and components. The materials and components shall be tested for the characteristics specified in table IV. The lot size shall be expressed in units of pounds for paranitrophenol, and gallons for all other materials. The sample unit for testing shall be one pound or quart, as applicable. The sample size shall be as shown in table III below. The sample unit shall be randomly selected throughout the lot. The component lot shall be unacceptable if one or more sample unit fails to meet any of the test requirements specified.

TABLE III. Sampling for component testing

Lot Size	Sample Size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE IV. Instructions for testing of components

Characteristic	Specification Reference		Requirements Applicable To		Number Determinations Per sample unit	Results Reported As	
	Requirement	Test Method	Individ. Unit	Lot Aver.		Pass or Fail	Numerically to nearest.
Paranitrophenol, percent	3.1.1	4.3.1	X		2		Average of 2 determinations, to nearest 0.1 percent.
Melting point, °C.	3.1.1	4.3.1.1	X		2		Average of 2 determinations to nearest °C.
Mineral oil Specific gravity, 20° /4°C.	3.1.2	402 1/	X		2		Average of 2 determinations, to nearest 0.001 value.
Color, (ASTM color)	3.1.2	102 1/	X		1		Pass or fail.
Neutralization number	3.1.2	5105 1/	X		2		Average 2 determinations, to nearest 0.01 value.
Viscosity, SU in seconds @ 100°F.	3.1.2	304 1/	X		2		Average of 2 determinations, to nearest second
Cyclohexanone Refractive index @ 25°C.	3.1.4	4.3.2	X		2		Average of 2 determinations, to nearest 0.0001 value.
Specific gravity, 25° /4°C.	3.1.4	402 1/	X		2		Average of 2 determinations, to nearest 0.001 value.
Color, (ASTM Color)	3.1.4	102 1/	X		1		Pass or fail.
Cyclohexanone Distillation range	3.1.4	4301 2/	X		2		Average of 2 determinations, to nearest 0.1° C.
Water content	3.1.4	4.3.3	X		2		Average of 2 determinations, to nearest 0.1 percent

1/ Denotes method applicable to Fed. Test Method Std. No. 791

2/ Denotes method applicable to Fed. Test Method Std. No. 141

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4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and AQLs set forth in 4.2.3.3. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in filled unit containers for examination in 4.2.2.2 and 4.2.2.3, in units of shipping container for examination in 4.2.3.1 and in units of pallets for examination in 4.2.3.2.

4.2.2.2 Examination of unit containers. The sample units for this examination shall be one filled unit container.

Examine	Defects
Container	Not type specified. Any evidence of leakage. Any unsound seam. Any break or dent in body of container. Rusty container. Closure not as specified.
Contents	Not homogeneous liquid.
Marking	Not in accordance with contract and specification requirements. Illegible, incorrect, or incomplete. Unit container not labeled as specified (see 3.3.1.1 and 3.3.1.2).

4.2.2.3 Examination of the filled unit containers for compliance with net contents. The end item shall be examined for defects in net contents. The sample unit for this examination shall be one filled unit container. The lot shall be acceptable if the average net content per container examined is less than the quantity specified.

4.2.3 Examination of preparation for delivery.

4.2.3.1 Examination for packaging, packing, and marking. An examination shall be made to determine that packaging, packing, and markings comply with the requirements of section 5 of this specification. The sample unit for this examination shall be one shipping container fully prepared for delivery, except that it shall not be palletized and need not be sealed. Closed shipping containers fully prepared for delivery that have not been palletized shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot. Defects shall be scored in accordance with the list below.

Examine	Defects
Packing	Not in accordance with contract requirements. Shipping container not as specified. Closure not accomplished by specified or required methods or materials. Any leakage of contents.
Marking (exterior or interior as applicable)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application. Not in accordance with contract requirements.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping, or stapling. Bulged or distorted container.

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4.2.3.2 Examination for palletization. An examination shall be made to determine that the palletization requirements comply with section 5 of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defects
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Wood cap not used over and under the load. Double tray cap (when applicable) not used. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; incomplete; or improper size, location, sequence, or method of application.

4.2.3.3 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.2.2.2	I	2.5
4.2.2.3	S-2	N/A
4.2.3.1	S-2	2.5
4.2.3.2	S-1	6.5

4.2.4 Testing of the end item. The end item shall be tested for the characteristics specified in table VI. The lot size shall be expressed in gallons of leather dressing compound. The sample unit for testing shall be a one pint composite obtained by combining equal portions from samples selected at random throughout the lot and thoroughly mixed to yield a homogeneous solution. The frequency shall be as indicated in table V below. The composite shall be placed in a clean, dry, sealed container with identifying lot and date. Care shall be exercised to prevent contamination or alteration of the composite during the sampling, compositing, storage, and testing. The lot shall be unacceptable if the composite fails to meet any test requirements specified.

TABLE V. Sampling for end item testing

Lot Size (gallons)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

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TABLE VI. Instructions for testing of end item

Characteristic	Specification reference		Requirements applicable to		Number Determinations Per sample unit	Results reported as	
	Requirement	Test Method	Samp. Unit	Comp-osite samp.		Pass or Fail	Numerically to nearest
Type I:							
Composition	Table I	1/					
Paranitrophenol	Table I	4.3.4		X	2		Average of 2 determinations, to nearest 0.1 percent.
Nonvolatile residue	Table I	4.3.5	X		2		Average of 2 determinations, to nearest 0.1 percent
Unsaponifiable matter	Table I	4.3.6		X	2		Average of 2 determinations, to nearest 0.1 percent
Specific gravity	Table I	4.3.7		X	2		Average of 2 determinations, to nearest 0.01 value.
Matter insoluble in alcohol	Table I	4.3.8		X	1		Pass or fail
Type II:							
Composition	Table II	1/					
Paranitrophenol	Table II	4.3.9		X	2		Average of 2 determinations, to nearest 0.1 percent
Water content	Table II	4.3.10	X		2		Average of 2 determinations, to nearest 0.1 percent.
Nonvolatile residue	Table II	4.3.5	X		2		Average of 2 determinations, to nearest 0.1 percent.
Organically combined SO_3	Table II	4.3.11		X	2		Average of 2 determinations, to nearest 0.1 percent.
Unsaponifiable nonvolatile matter	Table II	4.3.12		X	2		Average of 2 determinations, to nearest 0.1 percent.
Free fatty acid	Table II	4.3.13		X	2		Average of 2 determinations, to nearest 0.1 percent.
pH (acidity)	Table II	4.3.14		X	2		Average of 2 determinations, to nearest 0.1 pH unit.
Matter insoluble in water	Table II	4.3.15		X	1		Pass or fail.

1/ Certificate of compliance will be furnished to the contracting officer for composition requirements except for paranitrophenol content.

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4.3 Test procedures.

4.3.1 Paranitrophenol. Weigh 1.000 to 1.500 g. of the paranitrophenol to be analyzed and place in a kjeldahl digestion flask. Add 30 milliliters (ml.) of concentrated sulfuric acid (nitrogen free) and shake until thoroughly mixed. Allow to stand 30 minutes with frequent shaking or until complete solution results. Add 5 g. of crystallized sodium thiosulfate and let stand at least 10 minutes with frequent shaking. Warm contents gently, shake and let stand for another 5 minutes. Then heat slowly until foaming subsides and charring of all the material has occurred. Digest, distill, and titrate ammonia in accordance with method 6441 of Fed. Std. No. 311.

$$\text{Percent of paranitrophenol} = \frac{\text{milliliters of standard acid} \times \text{normality} \times 13.911}{\text{Weight of sample}}$$

4.3.1.1 Melting point. The melting point of the paranitrophenol shall be determined by the capillary tube method, using a Thiele-Dennis type melting tube, or equivalent, heated at a rate not to exceed 0.5° C. per minute. The temperature at which the sample becomes completely liquid shall be the melting point.

4.3.2 Refractive index. The refractive index shall be determined using either an Abbe refractometer or Zeiss-Butyro refractometer or equivalent.

4.3.3 Water content. The water content shall be determined in accordance with method 202 of Fed. Test Method Std. No. 536 except that a test specimen of 100.00 ± 0.01 g. of cyclohexanone and 100 ml. of toluene solvent shall be used. No barium chloride shall be used.

4.3.4 Paranitrophenol. Dissolve 1.00 ± 0.01 g. of the dressing compound in chloroform, transfer to a 100-ml. volumetric flask, make up to volume with chloroform and mix.

To each of nine 100-ml. volumetric flasks, add 20.0 ml. of a 10 percent solution of sodium citrate ($Na_3C_6H_5O_7.2H_2O$) in water. Use eight of these flasks to prepare standards containing the following increments of paranitrophenol: 0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, and 0.50 mg. by adding 0.5, 1, 1.5, 2, 2.5, 3, 4, and 5 ml., respectively, of a 0.01 percent solution of paranitrophenol prepared by dissolving 1.0 g. of pure paranitrophenol in 1 liter of 0.05 NaOH, and dilution 50.0 ml. of this solution to 500 ml. with distilled water.

Place 1 ml. of the chloroform solution of the dressing into the other volumetric flask. Treat the standards and unknown alike in the subsequent procedure. Reduce the paranitrophenol to paraaminophenol by adding slowly, with shaking, a freshly made solution of titanous chloride prepared by diluting 1 ml. of a 20 percent reagent grade solution to 100 ml. with boiled and cooled distilled water until the color changes from yellow to slight purple (2.5 to 10 ml. will be required). Shake vigorously. Add 30 ml. of a borate buffer solution containing 30 g. of borax ($Na_2B_4O_7 10H_2O$) reagent grade per liter of distilled water and adjust to a pH of 9.6 with a 10 percent solution of sodium hydroxide. Add 10 ml. of a 5 percent solution of phenol in distilled water and again shake vigorously. Make to volume with the borax buffer solution and mix thoroughly. Let stand from one to three hours in subdued light to develop color. If a cloudy color develops in the unknown, filter this through a fritted Gooch crucible of medium porosity before the color is read.

Determine the milligrams paranitrophenol in the unknown colorimetrically by comparison with the standards by one of the following two methods, depending on the type of equipment available:

1. With a spectrophotometer.

Read the colors of the standards at 630 millicrons and plot the curve. Read the unknown and obtain milligrams of paranitrophenol from the standard curve.

2. With an electrophotometer.

Read the color of standards and unknown using a filter showing maximum transmission at about 630 millimicrons (Fisher's electrophotometer with 650-A filter or equivalent is satisfactory). Plot a standard curve and determine the concentration of paranitrophenol of the unknown as in 1.

Percent paranitrophenol in the compound = Milligrams of paranitrophenol found in the unknown × 10.

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4.3.5 Nonvolatile residue. Weigh 10.0 g. of the compound and place in a tared crystallizing dish approximately 6 centimeters in diameter. The specimen shall be dried to constant weight in a vacuum oven at a temperature of 70° ± 5°C. and at a pressure of 20 to 30 millimeters of mercury. Constant weight shall be assumed when successive dryings for 1 hour show losses of less than 0.05 percent. The dish and residue shall be cooled in a desiccator and weighed. Calculate the percentage of nonvolatile material as follows:

$$\text{Percent nonvolatile material} = \frac{\text{Weight of residue}}{\text{Weight of sample}} \times 100$$

4.3.6 Unsaponifiable matter. Unsaponifiable matter shall be determined in accordance with method 6331 of Fed. Std. No. 311, except that 3.00 ± 0.01 g. of leather dressing compound shall be used.

4.3.7 Specific gravity. The specific gravity shall be determined in accordance with method 401 of Fed. Test Method Std. No. 791.

4.3.8 Insoluble matter in alcohol. Weigh a 5.00 ± 0.01 g. sample and digest with 200 ml. of freshly boiled ethyl alcohol (95 percent or higher and neutral to phenolphthalein) in a covered vessel on a steam bath until the leather dressing compound is dissolved. Filter through a weighed Gooch crucible with suction, protecting the solution from carbon dioxide and other acid fumes during the operation by covering with a watchglass. Wash the residue in the crucible with hot neutral alcohol until free from leather dressing compound. Dry the crucible with residue at 105° ± 2°C. for three hours, cool, and weigh. Determine the total matter insoluble in alcohol.

4.3.9 Paranitrophenol. Dissolve 1.00 ± 0.01 g. of the dressing compound in chloroform and transfer to a 200-ml. volumetric flask, make up to volume with chloroform, and mix. Proceed as specified in 4.3.4. Express results in percentage of paranitrophenol based on the weight of the original specimen as follows:

The milligrams of paranitrophenol determined in the unknown X 20 = percent paranitrophenol in the compound.

4.3.10. Water content. The water content shall be determined in accordance with method 202 of Fed. Test Method Std. No. 536, except that a test specimen of 15.00 ± 0.01 g. of leather dressing compound shall be used. Use 80 g. of xylene as solvent and 30 g. of oleic acid to prevent foaming and jellying.

4.3.11 Organically combined SO_3.

4.3.11.1 Reagents.

Ethyl ether, U.S.P.
Sodium chloride - Solid, c.p. NaCl.
Sodium sulfate - Anhydrous, c. p. Na_2SO_4.
Sulfuric acid, approximately 1 N.
Sodium hydroxide, approximately 0.5 N and 1 N.
Methyl orange indicator - 0.1 percent water solution of methyl orange.

4.3.11.2 Procedure. Weigh 8.00 ± 0.01 g. of the sample, transfer into a 250-ml. separatory funnel containing 50 ml. of saturated NaCl solution at 25°C., 5 drops of methyl orange indicator, and 50 ml. of ether. Shake the mixture and neutralize with 1 N H_2SO_4 until the lower layer is distinctly pink (about 0.2 ml. in excess). At this point, some oils may form three layers instead of two. If this happens use a fat solvent consisting of a mixture of two parts of ether and one of alcohol. Allow the mixture in the separatory funnel to settle for at least 5 minutes, draw off the lower layer into another separatory funnel and wash by shaking the ether layer with 25 ml. portions of a saturated NaCl solution until practically neutral to methyl orange, that is, until 1 drop of 0.5 N NaOH turns the wash water alkaline to methyl orange. At this point make 3 additional washings of the ether layer with 25 ml. portions of NaCl solution. Allow all separations to settle for at least 5 minutes. Combine the water layers, and extract with two 25 ml. portions of ether. Combine the last two ether extractions and wash with NaCl solution until free from acid, as in the case of the ether layer in the first funnel. Combine the ether layers in the first funnel. Carefully remove any water that may settle and dehydrate the ether layer as follows: Add 5 g. of anhydrous Na_2SO_4, shake vigorously for 5 minutes, and filter directly into a 150-ml. beaker, placed in a bath of warm water. Wash the separatory funnel and filter with ether until free from fat (absence of oil stains on the filter paper after drying) and add the filtrate to the beaker. To avoid creeping of the oil, the volume in the beaker should at no time during the filtering and washing exceed 50 ml. (one-third full). Evaporate the ether solution

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until the volume has been reduced to about 20 ml. and transfer the residue to a tared 50 ml. crucible (high form). Immerse the crucible in a 100-ml. beaker containing warm water until practically all of the ether has evaporated. Rinse the beaker with two 10 ml. and three 5 ml. portions of ether, or until all of the oil has been transferred to the crucible; allow each portion of rinsing ether to evaporate before the next rinsing is made. Burn the solvent-free residue gently, and ignite to a dull red heat to constant weight. To prevent creeping of the oil and to hasten evaporation, stir the solution with a glass rod; before the oil is burned, wipe the rod clean with ashless filter paper and add the paper to the crucible. To oxidize traces of carbon or sodium sulfide that might form, moisten the ash with hydrogen peroxide (30 percent) and again carefully ignite to constant weight. Calculate the percentage of ash as follows:

$$\text{Extracted ash, percent} = \frac{\text{Wt. of ash in g.}}{\text{Wt. of sample in g.}} \times 100$$

Calculate the percentage of organically combined SO_3, as follows:

Organically combined SO_3, percent = 1.1267 X extracted ash, percent.
Where 1.1267 is the molecular ratio of $2SO_3$: Na_2SO_4.

4.3.12 Unsaponifiable matter.

4.3.12.1 Apparatus.

300-ml. Erlenmeyer flask and glass stopper, made of heat-resistant glass. Condenser consisting of a glass tube, 915 mm. (36 in.) in length, and 8mm. (5/16 in.) in outside diameter. The lower end of the glass tube shall be flared and ground to fit the mouth of the Erlenmeyer flask.

4.3.12.2 Reagents.

Sodium chloride - Solid, c.p. NaCl.
Potassium hydroxide - Approximately
0.5 N aqueous KOH solution.
Alcoholic potassium hydroxide - Approximately but not less than
0.5 N alcoholic KOH solution
Ethyl ether, U.S.P.
Sulfuric acid - Approximately 1.0 N.
Sodium hydroxide - Approximately 1.0 N.
Ethyl alcohol - U. S. P. 95 percent.
Methyl orange indicator - A 0.1 percent water solution of methyl orange.

4.3.12.3. Procedure. Weigh 10.00 ± 0.01g. of the sample, transfer into the Erlenmeyer flask and boil under the air condenser with 1 N H_2SO_4 for 1-1/2 hours or until both the oil and water layers are perfectly clear. Use glass beads to prevent bumping. The volume of H_2SO_4 added shall be sufficient to neutralize the total alkalinity of the specimen plus 25 ml. in excess. Regulate the heating so that the solution boils rather vigorously but with very little evaporation taking place. At the end of the heating period allow the contents to cool, wash the condenser with a spray of water from a wash bottle and disconnect the condenser. Add 30 g. of NaCl, 25 ml. of ether, 50 ml. of water, and five drops of indicator and titrate the solution with 1.0 N NaOH. During the titration, stopper the flask frequently and shake the contents of the flask thoroughly. After cooling, transfer the titrate solution to a 250 ml. separatory funnel and shake with 50 ml. of ether. Draw off the water layer into another separatory funnel and extract twice with 25 ml. portions of ether. Wash the combined ether layers with 15 ml. portions of saturated salt water until the wash water is neutral to methyl orange. Transfer the ether layer to a 150-ml. beaker and evaporate to dryness. The dried mass is then dissolved in a 50 ml. mixture of 50 percent alcohol and 50 percent water and transferred to a separatory funnel. Rinse the beaker with a like 50 percent alcohol mixture, add to the funnel and extract this with an ether mixture consisting of 3 parts of petroleum ether and 1 part of ethyl ether, using three consecutive portions of the ether mix. Discard the alcohol water portion which contains the undecomposed sulfonates. Transfer combined ether extractions to a tared 125 ml. assay flask. Dry to constant weight and calculate total desulfated fatty matter as follows:

$$\text{Total desulfated fatty matter, percent} = \frac{\text{Wt. of residue in g.}}{\text{Wt. of sample in g.}} \times 100$$

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Weigh 2.00 ± 0.01 g. of above dried fatty matter, transfer into the 300-ml. Erlenmeyer flask and add 25 ml. of the alcoholic KOH. Attach the condenser and simmer the contents (without loss of alcohol) for 1 hr. with occasional swirling over an electric hot plate or other source of heat. Transfer the contents of the flask to a 250 ml. separatory funnel and wash the flask several times with a total of 50 ml. of water, pouring it into the separatory funnel. Extract the solution while still warm (about 30°C.) with 50 ml. of ether (rinse the saponification flask with the ether before adding it to the separatory funnel), shaking vigorously for about 1 minute, and allow the layers to settle and clear. Draw off the lower layer in a second 250-ml. separatory funnel and extract in a similar manner with two 50 ml. portions of ether. Add 20 ml. of water to the combined ether layers, turn the separatory funnel and contents over gently about six times and allow the layers to settle and clear. Draw off the lower layer and discard. Wash the ether layer twice more with 20 ml. portions of water, shaking vigorously after each addition. Add 20 ml. of 0.5 N KOH to the washed ether layer, turn the separatory funnel and contents over gently about six times, and allow the layers to settle and clear. Draw off the lower layer and discard. Wash the ether layer with 20 ml. of water, shaking vigorously. Continue washing with 20 ml. of 0.5 N KOH followed by 20 ml. of water, shaking vigorously after each addition, until the alkali layer upon strong acidification with HCl and settling for a few minutes is only faintly opalescent. Finally wash the ether layer with 20 ml. portions of water until the wash water is no longer pink to phenolphthalein. Transfer the ether layer to a tared 150-ml. beaker and evaporate over a hot-water bath until practically free of ether. Heat the beaker at 80° ± 2°C. until constant weight is obtained. Calculate the unsaponifiable matter as follows:

$$U = \frac{R}{G} \times P$$

Where: U = unsaponifiable matter, percent.
R = weight of residue in g.
G = weight of desulfated fatty matter in g.
P = Percentage of desulfated fatty matter.

4.3.13 Free fatty acids (as oleic).

4.3.13.1 Reagents.

Sodium hydroxide, 0.5 N.
Phenolphthalein indicator - 1 percent solution in 95 percent ethanol.
Ethyl ether, U.S.P.
Ethanol, 95 percent - A.C.S.

4.3.13.2 Procedure. Dissolve 10.00 ± 0.01 g. of the sample in a mixture of 50 ml. neutral alcohol and 25 ml. of ether, add five drops of phenolphthalein indicator, and titrate the solution with 0.5 N NaOH until the pink color persists after vigorous shaking. Calculate the free fatty acids in percent by weight as follows:

$$\text{Percent free fatty acids} = \frac{\text{ml. of alkali} \times N \times 28.25}{\text{Weight of sample}}$$

Paranitrophenol will be included as oleic acid in this test. Correct for the paranitrophenol acidity as follows:

True percent of free fatty acids (as oleic) = Percent free fatty acids (as oleic) found - percent paranitrophenol in dressing × 2.03.
The percent paranitrophenol in dressing may be determined as specified in 4.3.4.

4.3.14 pH (acidity). The ph of a 10 percent emulsion shall be determined in accordance with method 6621 of Fed. Std. No. 311.

4.3.15 Insoluble matter in water. Weigh a 5.00 ± 0.01 g. sample and digest with 200 ml. of water in a covered vessel at 60°C. until the leather dressing compound is dissolved. Filter through a weighed Gooch crucible with suction, protecting the solution from carbon dioxide and other acid fumes during the operation by covering with a watchglass. Wash the residue in the crucible with warm water. Dry the crucible with residue at 105° ± 2°C. for three hours, cool and weigh. Determine the total matter insoluble in water.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. The leather dressing compound shall be packaged in 1-pint, 1-quart, 1-gallon, or 5-gallon quantities, as specified (see 6.2).

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5.1.1.1 One-pint, 1-quart, and 1-gallon quantities. One-pint, 1-quart, or 1-gallon of the leather dressing compound shall be filled into oblong metal cans conforming to type V, class 4, with inner seal, and fabricated from 1.25-pound hot dipped tinplate of PPP-C-96. The filled one-pint cans shall be packaged in accordance with the level A packaging requirements in the appendix of PPP-C-96.

5.1.1.2 Five-gallon quantity. Five gallons of the leather dressing compound shall be filled into steel pails, with interior coating, conforming to type I, class 3 of PPP-P-704. Screw cap closures and outer seals shall be furnished on all pails.

5.1.2 Level C. Leather dressing compound shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Leather dressing compound, packaged in 1-pint, 1-quart, and 1-gallon cans as specified in 5.1, shall be packed in fiberboard or cleated plywood shipping containers, as specified (see 6.2) in accordance with level A packing requirements in the appendix of PPP-C-96. Five-gallon pails shall not require overpacking.

5.2.2 Level B. Leather dressing compound, packaged in 1-pint, 1-quart, and 1-gallon cans as specified in 5.1, shall be packed in fiberboard shipping containers in accordance with the Level B packing requirements in the appendix of PPP-C-96. Five-gallon pails shall not require overpacking.

5.2.3 Level C. Leather dressing compound, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization.

5.3.1 Civilian agencies. When specified (see 6.2), leather dressing packed as specified shall be palletized in accordance with 5.3.2, 5.3.2.1, and 5.3.2.2.

5.3.2 Military requirements. Unless otherwise specified (see 6.2), leather dressing packed as specified shall be palletized in accordance with MIL-STD-147.

5.3.2.1 One-pint, one-quart, and one-gallon cans. One-pint, one-quart, and one-gallon cans of leather dressing shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.3.2.2 Five-gallon pails. Five-gallon pails of leather dressing shall be palletized in accordance with Load Type III of MIL-STD-147. The load shall first be bonded with a horizontal strap around each course except top in accordance with bonding means E. The primary and secondard straps shall be applied in accordance with bonding means K and L. A wood cap shall be positioned over and under the load in accordance with storage aid 5.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The leather dressing compounds covered by this specification are intended to provide resistance to cracking and mildew, preserve the original qualities of the leather and improve the qualities of leather items which have been in use. The leather dressing compounds (Types I and II) are intended for treating leather objects which come into occasional contact with the skin. When boots and shoes are treated, socks must be worn.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) The type required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Size package required (see 5.1.1).
(e) Type of shipping container desired for level A packing (see 5.2.1).
(f) When palletization is required for civilian agencies (see 5.3.1).
(g) When palletization is not required for military procurements (see 5.3.2).

6.3 Sulfonated cod oil. Satisfactory results have been obtained by utilizing a grade of sulfonated cod oil conforming to the following general requirements:

	Minimum	Maximum
Water, percent	24.0	26.0
Unsaponifiable matter, percent	----	1.2
Organically combined SO_3, percent	3.5	5.5
Free fatty acid, as oleic, percent	----	25.0
Ash, percent	----	7.5
Total alkali, percent	----	1.5
Stability of 10 percent emulsion in water, hours	24.0	----
pH, 10 percent solution	5.5	6.5

6.4 Sulfonated neatsfoot oil. Satisfactory results have been obtained by utilizing a grade of sulfonated neatsfoot oil conforming to Type II of C-N-200.

MILITARY CUSTODIANS:

Army - GL
Air Force - 11

Review activities:

Army GL,MD,ME, MR, MU
Air Force - 11, 84

User activities:

Navy - MC, SH

Preparing activity:

Army-GL

CIVIL AGENCY INTEREST:

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